

Cora users guide

version 1.7 (V7.6.9)

draft

Cora users guide

version 1.7 (V7.6.9)

dr. E.J. Spee

1206709-000

Title

Cora users guide

Client

Shell

Project

1206709-000

Reference

1206709-000-GEO-0075-
gbh

Pages

33

Keywords

Probabilistic analysis

Summary

The CORA enables the engineer to create reliability analysis for complex models using advanced probabilistic techniques. This manual describes the input and working method. It is elaborated for the method called Geertsma. It includes the method to insert a used defined model.

References

Shell contract PT25399 Variation no. 5, date 17.02.2015

Deltares offer 1206709-000-GEO-0066-gbh dd. 6 February 2015

Version	Date	Author	Initials	Review	Initials	Approval	Initials
	jan. 2016	dr. E.J. Spee		dr.ir. P. Hölscher		ing. A.T. Aantjes	
				dr. ir. R. Brinkman			

State

draft

This is a draft report, intended for discussion purposes only. No part of this report may be relied upon by either principals or third parties.

Contents

1 Introduction: what can be calculated with Cora	1
2 Installation	3
3 First time using Cora	5
4 Overview of a calculation	7
5 Failure definition management	9
6 Geertsma	13
6.1 General settings	13
6.2 Model Parameters	13
6.3 Parameters for the defined failures	14
6.3.1 Correlations	15
6.3.2 Navigation buttons	16
7 Calculations with other models	17
7.1 Quickblocks	17
7.2 GeoMec	17
7.3 User-defined model	17
8 Scenario results	19
8.1 Controls	19
8.2 Creating scenario table	20
8.3 Creating proxy table	21
9 Scenario plots	23
9.1 Tornado chart	24
9.2 True solution chart	25
9.3 Coefficients chart	26
9.4 Navigation buttons	26
10 Reliability	27
10.1 Current selected engine	27
10.2 Probabilistic configuration settings	28
10.3 Run engine	29
10.4 Run proxy	29
10.5 Probabilistic method table	30
11 Reliability results	31
11.1 Navigation buttons	31
12 Settings	33
12.1 Navigation buttons	33

Appendices

A Working with a user defined model

A.1 Environment

A.2 Example 1

A.3 Example 2

A-1

A-1

A-2

A-2

1 Introduction: what can be calculated with Cora

Cora is an interactive program for probabilistic analysis with geo-mechanical models.

The following models are implemented:

- Geertsma.
- Quickblocks.
- Geomec.
- User defined model.

This manual applies to version 1.7. It treats the general input of the Cora software, the input for using Geertsma and the user defined model.

The background of the probabilistic methods is described in the report “CORA Design” report Deltares 1206709-0000-GEO-0074 January 2016. The usage of the models that are implemented is described in the manual of the model itself. More information on the coupling with the Geomec model is described in the report “Design interface CORA Geomec” report Deltares 1206709-000-GEO-0010.

2 Installation

The user interface of Cora and the behind calculation engine are separated. This user interface is accessed through an Excel document, named "Cora users guide.xlms". It runs the best using Microsoft Excel 2007 or higher, that it provides an optimum user experience. Excel should be default installed on your Shell laptop or can be installed via **Add GI Software**.

Installation of the Cora software requires the following steps:

- Unpack the zip file into a folder on the hard drive using the password Simon32. The **C:\Apps folder** is the recommended location.
- Launch the installation by running the MSI installation file and follow the installation steps. The **C:\Apps\Cora** is the recommended installation folder location. Create this folder if it not exists.

After a successful installation of the Cora software both the calculation kernel and Excel sheet are installed on your laptop. Note that after a successful installation an Excel sheet named Cora can be found in the installation folder, this should be **C:\Apps\CoralCora.xlsm**.

3 First time using Cora

The Cora Excel sheet uses macro's, so when you opening the Excel sheet for the first time Excel will give a warning about this. Cora do not contains any harmful macro's so please click enable macro's to continue.



Figure 3.1 Switch on macros in Excel

When macros' are enabled you will get the following warning messages, please click OK.

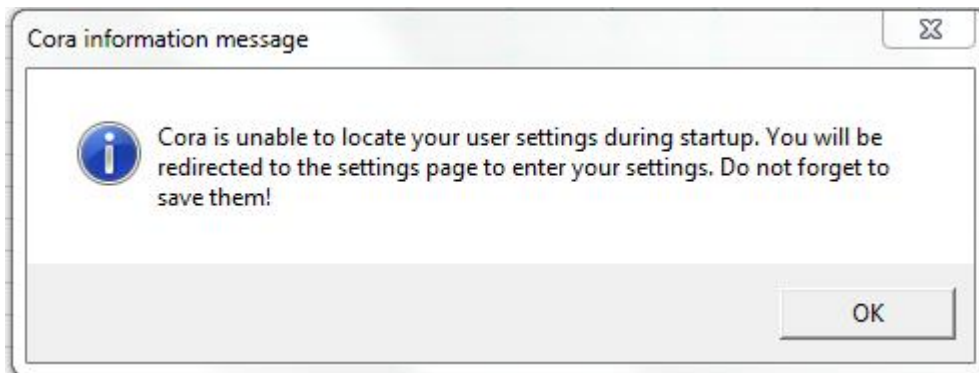


Figure 3.2 Information message to be ignored

Please enter the following path settings on the settings screen and **do not forget** to save them by clicking "Save settings".

			Save settings
Sheet	Key	Value	
Cora	Path cora application	C:\Program Files (x86)\Deltares\CORA\Binaries	
QuickBlocks	Python path	C:\Program Files\QuickBlocks 2.1\python.exe	
QuickBlocks	QuickBlocks path	C:\Program Files\QuickBlocks 2.1\	
Cora	Verbose		2
GEOMECH	GEOMECH path	C:\Program Files\GEOMECH 4.2 - SVS	
GEOMECH	SaveAllCoralInput	TRUE	
GEOMECH	SaveAllModelResults	TRUE	
GEOMECH	SaveModelResultsFailedCases	FALSE	
GEOMECH	SaveInputFilesFailedCases	TRUE	

Figure 3.3 Settings needed at installation

Explanation of settings:

- Key Path Cora application, this is the path where Cora is actually installed. This should be C:\Program Files (x86)\Deltares\CORA\Binaries.
- Key Python path, this path should point to your Python installation and is only required when you want to use Quickblocks. This should be C:\Python\python.exe.
- Key QuickBlocks path, this should point to your QuickBlocks installation folder and is only required when you're using QuickBlocks. This should be C:\Apps\QuickBlocks
- Key Verbose, this value determines the amount of log information, please enter a value between 1 and 3.
- Key GEOMECH path, this should point to your GEOMECH installation folder and is only required when you're using GEOMECH. This should be C:\Program Files\GEOMECH 4.2 – SVS.

When your settings are saved, Cora is ready to use. Please navigate to the landing sheet called "Home".

4 Overview of a calculation

A Cora simulation is built up by a logical sequence of steps. In this manual we assume that you have a problem that is already solved by one of the models that is implemented in Cora. This solution is deterministic. It can be considered as a best estimate solution, or maybe a reasonable upper limit, depending on the parameters you used. Cora will help you to calculate the probability of failure of the problem under consideration. Here failure is defined in general terms: a system fails if it doesn't fulfil the requirements.

The following steps are done within Cora

- Definition of the types of failure that should be taken into account.
- A scenario study that helps you to decide which parameters must be involved in the final reliability calculation.
- A reliability study.

The two last steps have always two parts: the definition of the problem before the calculation and the (graphical) presentation of the results after the calculation.

5 Failure definition management

This sheet provides functionality to add, delete or modify your definition of failure.

Type	Failure Name	Failure mechanisms	Engine	Input file	Calculation modus
Fault	f1	Shear failure	Geertsma	na	na

Failure definition

Engine

Engine:

Input File:

Calculation modus:

Failure

Failure type:

Failure mechanisms:

Failure name:

< 2 of 2 >

Add Delete Update Done

Home Scenario results Scenario plots Reliability Reliability results Settings CSV Viewer

To open the failure definition form, please click the “Failure definition” button on the top left. There are **five steps required** to add a new failure type, these are the required steps:

- Select an engine from the dropdown box right of the label called engine. In this version there are for engine available, Geertsma, QuickBlocks, GEOMECH and USER model. For the last two engines you also need to specify an input file and for GEOMECH also the calculation mode. **(For example Geertsma)**.
- Select a failure type from the dropdown box right of the label called Failure type. Failure types in this dropdown box are filtered on the selected engine the previous step **(For example Fault)**.
- Select a failure mechanism from the dropdown box right of the label called failure mechanism. Failure mechanisms are filtered on the selected failure type in the previous step **(For example shear failure)**.
- Enter a failure name in the textbox right from the label Failure name. This name should be unique and **only** contains alphanumeric values. **(For example F1)**.

- Finally the defined failure type is added by clicking the “Add” button. When there is no validation warning you should see that your failure type is added to the Excel sheet behind the form.

For every engine where failure types are added to, a button with the name of the engine is added to the top yellow area. From this button you can navigate to the specific input sheet for that engine (**In this example a button called “Geertsma”**).

There are 7 buttons located on the Failure definition form; these buttons offer the following functionality:



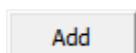
Button to navigate to your case file of the selected engine. (Only for UserModel and GEOMECH)



Button to load the previous failure.



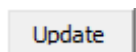
Button to load next failure.



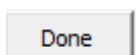
Button to finally add your defined failure to the collection of failure types, **input will be validated**.



Button to delete the current selected failure type.



Button to update the current selected failure type, **input will be validated**.



Button to close the failure definition form.

Table of available failure mechanisms for every failure type and engine. The available Failure types and failure mechanisms for GEOMECH and UserModel will depend on the selected case file.

Engine	Failure Type	Failure mechanism	Description
Geertsma	Fault	Shear failure	
Geertsma	Formation	Shear failure	
Geertsma	Formation	Tensile failure	
Geertsma	Subsidence	Subsidence	
Geertsma	Well	Axial compression	
Geertsma	Well	Axial elongation	
Geertsma	Well	Shear failure	
QuickBlocks	Fault	Shear failure	
QuickBlocks	Formation	Shear failure	
QuickBlocks	Formation	Tensile failure	
QuickBlocks	Formation	Compression failure	
QuickBlocks	Formation	Extension failure	
QuickBlocks	Formation	Compaction failure	
QuickBlocks	Subsidence	Subsidence	
QuickBlocks	Well	Axial compression	
QuickBlocks	Well	Axial elongation	

6 Geertsma

If you choose one or more failure mechanisms for the engine Geertsma, an additional sheet is generated for this engine. This sheet is meant to define your Geertsma input and is divided into three sections, general configuration section on the top, model base case in the middle and on the bottom all the failure types that are defined for this engine. All of these sections will be described in more detail in this section.

6.1 General settings

Geertsma Evaluation	
Model configuration	
Project output folder name	Example1
Max. Parallel runs	1

In this general configuration section you will configure two things, the project output folder and the maximum number of parallel runs that the calculation engine will use for this engine, default 1. The project output folder name needs to be unique; this folder will be created inside the root folder of the Cora Excel sheet location. So for example we fill in for the project output folder **Example1** and we're running the Cora Excel sheet from C:\Apps\Cora\ the folder Example1 will be created in C:\Apps\Cora\Example1.



This button will open windows explorer with your project output folder.

6.2 Model Parameters

In the model base section you will define your model parameter values and there uncertainties, by default all base parameter values are filled in. The following steps are required to define an uncertainty for a parameter:

- Select a parameter by clicking the green cell named Select and choosing an x value from the dropdown list, **for Example Reservoir Depth.**
- Select the distribution type for this parameter from the dropdown box, **for this example we select a Normal distribution.**
- Fill in the deviation value for this parameter, the mean value is always automatically filled in from the base case value. **For this example we enter a deviation of 500.**
- Additionally you can create an array for different mean values. This will let the calculation engine loop over different mean values for a single parameter. **For this example a Min value of 2000, a Max value of 2005 and a step size of 1.**

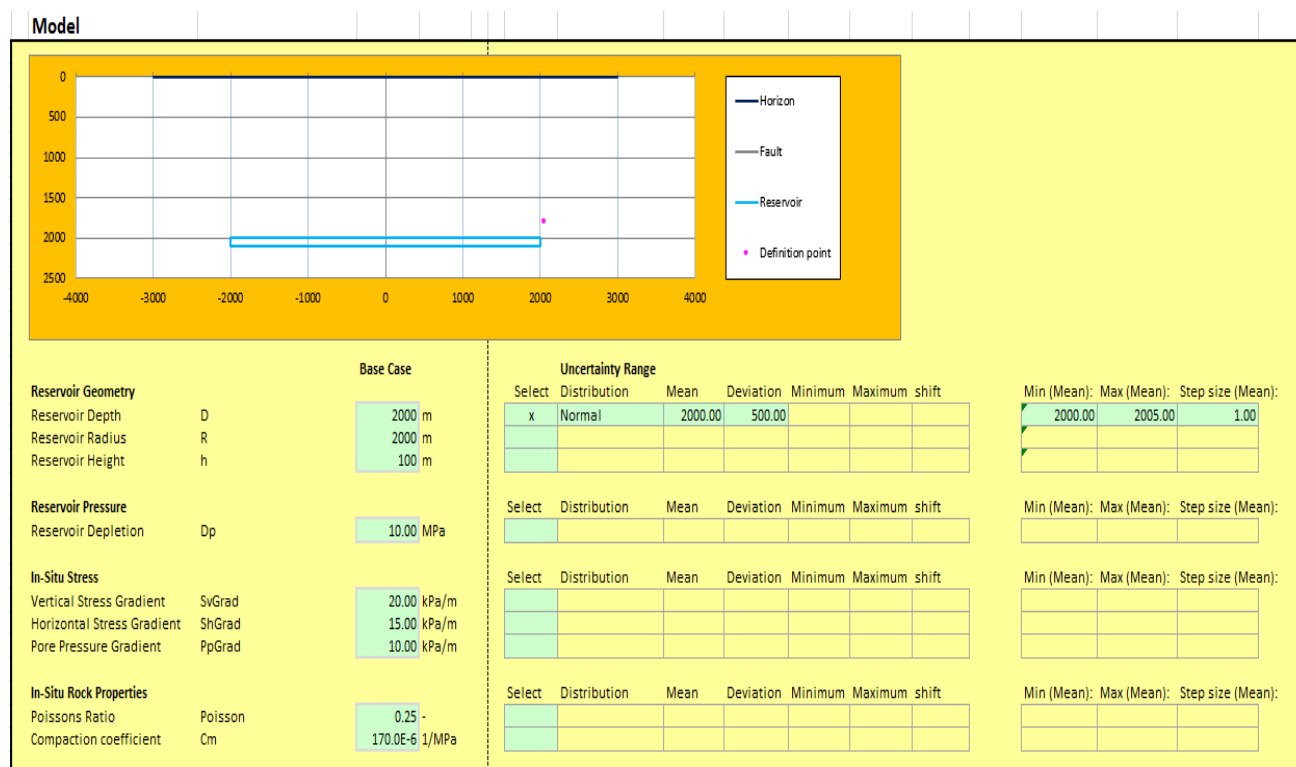


Table of available distribution types and required input:

Distribution	Required input	Description
Normal	Mean and deviation	well known Gauss distribution
LogNormal	Mean, deviation and shift	Logarithm of a normal distribution. Shift gives the lower bound of the distribution
Uniform	Minimum and maximum	
Deterministic	Mean	
TruncatedNormal	Mean, deviation, minimum and maximum	

6.3 Parameters for the defined failures

On the bottom section you will find a list of all failure that is added from the Cora sheet for this engine. Each failure is defined by one or more parameters, here called components. Every component has a number of parameters. The mean value is taken from the base case; in this section you define the uncertainty for each parameter, just like the above model parameters. In order to define uncertainty click on the green select box and choosing an "x" from the drop down list. Choose the distribution and fill in the required parameters (green boxes).

In the example we have added two uncertainties for this Fault with Shear failure: one on the defined cohesion and one on the angle of internal friction. As you can see there is only one value used for the mean value. All other parameters for this shear fault are treated as fixed values (deterministic).

Fault (Shear failure)			X									
Geometry												
Definition point (start or center)		Start	Select	Distribution	Mean	Deviation	Minimum	Maximum	shift	Min (Mean):	Max (Mean):	Step size (Mean):
Origin, r-coordinate	f1_point_r	2050 m										
Origin, z-coordinate	f1_point_z	1800 m										
Fault dip	f1_dip	40 deg										
Fault Length	f1_length	500 m										
Number of segments	f1_seg	100 Segments										
Properties Shear failure			Select	Distribution	Mean	Deviation	Minimum	Maximum	shift	Min (Mean):	Max (Mean):	Step size (Mean):
	f1_Coh	3.00	x	Normal	3.00	1.00				3.00	3.00	3.00
	f1_Phi	16.00	x	Normal	16.00	4.00				16.00	16.00	16.00

Default all components are selected and will be taken into account during the calculation. Each component can be deselected by removing the “X” value in the top right corner of the component.

6.3.1 Correlations

Cora is able to take into account the correlation between the selected parameters and components. The correlation form will give you a matrix representation of all selected parameters for that active engine at the moment. You can enter a value of -1 to 1 meaning the correlation between two parameters. The value 1 means fully correlated, the value -1 means fully anti-correlated. For example, in the correlation matrix you can see that f1_coh and D are fully correlated to each other. **Default** all parameters are not correlated to each other and set to 0.

Correlation Matrix

Uncertaintie properties: D f1_Coh f1_Phi

D	1	1	1
f1_Coh	0	1	1
f1_Phi	0	0	1

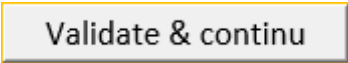
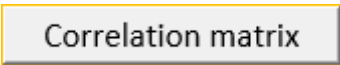
Ok

Definition of correlation table.

Value	Description
-1	fully anti-correlated
0	Not correlated
1	fully correlated

6.3.2 Navigation buttons

There are 3 buttons located in the top yellow area on every engine input sheet, this buttons will offer the following functionality:

	Button to navigate to the sheet named "Home".
	Button to validate the input and navigate to the "Scenario results" sheet.
	Button to open the "Correlation Matrix" form.

7 Calculations with other models

7.1 Quickblocks

It is also possible to make a calculation with Quickblocks. All available failure mechanisms and other information are stored in the Cora spread sheet.

7.2 GeoMec

Information on the coupling with Geomec is described in the report “Design interface CORA Geomec” report Deltares 1206709-000-GEO-0010.

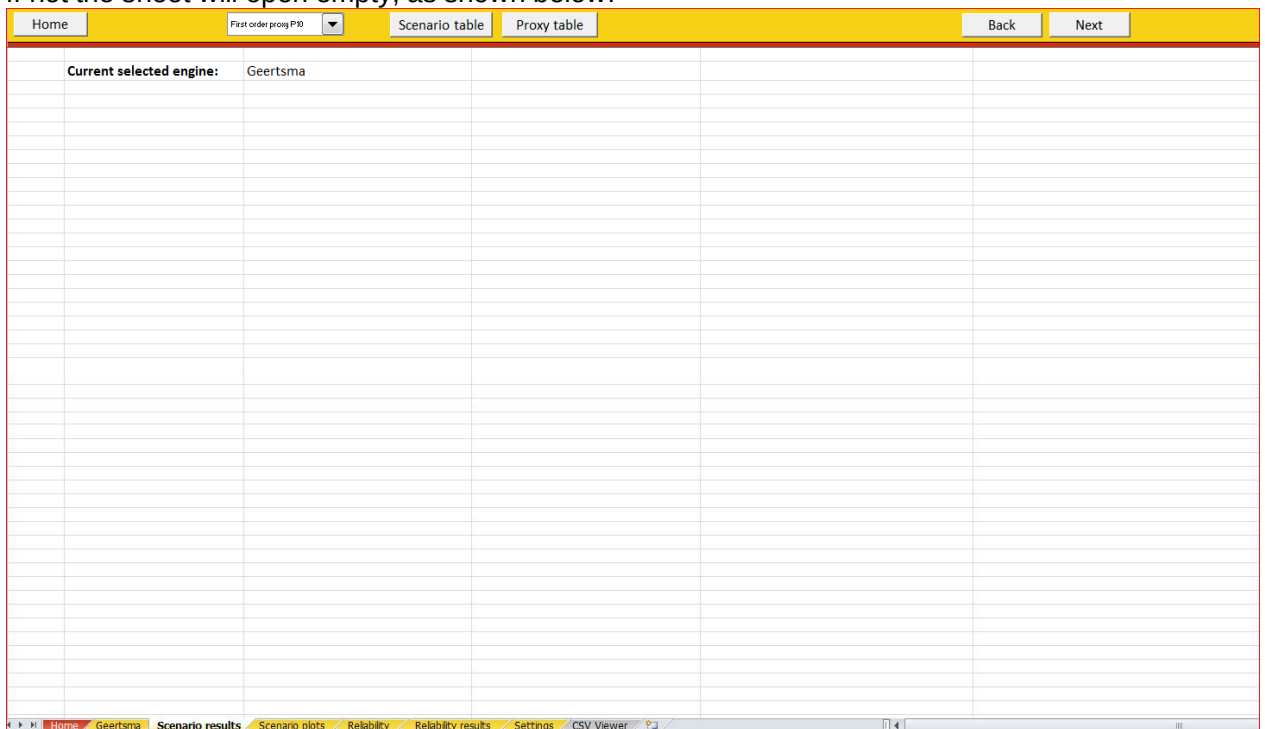
7.3 User-defined model

It is possible to add your own model to Cora, the so-called user defined model. It is strongly advised to get some user experience with Cora before. The way a user defined model is used is discussed in Appendix A.

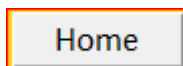
8 Scenario results

The next step in the evaluation is the calculation of the scenarios.

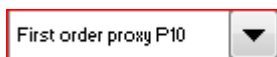
There are two ways to navigate to the “Scenario results” sheet. The first one is just to click the sheet tab. The second one is to click the “Validate & continue” button from the current active Engine. Both ways will open the “Scenario results” sheet showed in the below image. If there are previous calculation results for the selected engine, they will load and displayed as tables. If not the sheet will open empty, as shown below.



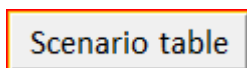
8.1 Controls



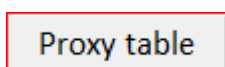
Button to navigate to the sheet named “Home”.



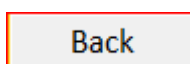
Drop down box to set the calculation order of the Proxy table.



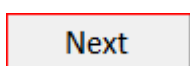
Button to generate the Scenario table.



Button to generate the Proxy table.



Button to navigate to the previous sheet.



Button to navigate to the next sheet.

8.2 Creating scenario table

To create a scenario table your first have to select the type¹ of proxy table. On the top of the sheet there is dropdown box where you can change the type of proxy, in the below table you can see the available proxy tables, default “First order proxy P10” is selected:

Proxy type
First order proxy P10
First order proxy P90
First order double proxy
Second order proxy
Experimental design

To build a scenario table, press the “Scenario table” button on the top of the sheet. This will start the calculation engine of Cora and will show a progress window to the user with a time estimate. After a successful calculation the progress window will close automatically. Notepad will open showing the log information and the actual scenario table will be written to the “Scenario results” sheet. When an error occurred during the calculation an error message will popup, after clicking the ok button Notepad will be automatically opened with the error log. The amount of detail showed in the log file can be changed in the “Settings” sheet by changing the verbose value. Below an example of a successful scenario table is showed

The screenshot shows the Deltares software interface. At the top, there is a yellow header bar with buttons for 'Home', 'First order proxy P10' (dropdown), 'Scenario table', 'Proxy table', 'Back', and 'Next'. Below the header, the 'Scenario table' sheet is displayed. It shows the 'Current selected engine' as 'Geertsma' and the 'Scenario-FirstOrderProxy10' table. The table has columns for 'D', 't1_Coh', 't1_Phi', and a numerical value. The data rows are highlighted in red. A Notepad window titled 'Geertsma_ProjectOutput 2014-09-04 16-05-52.txt - Notepad' is open in the foreground, showing log information: 'Info: *** START BUILD SCENARIO ***' and 'Info: *** END BUILD SCENARIO ***'. The bottom status bar shows tabs for 'Home', 'Geertsma', 'Scenario results', 'Scenario plots', 'Reliability', 'Reliability results', 'Settings', and 'CSV Viewer'.

D	t1_Coh	t1_Phi	
P-10-50-50	1359.224021	3	16
P-50-10-50	2000	1.718448061	16
P-50-50-10	2000	3	10.87379225
P-50-50-50	2000	3	16

The scenario table shows all combinations of the values of the parameters and components for the mean and characteristic values. These depend on the chosen type of proxy table.

¹ See the design document for background on the several proxy types.

8.3 Creating proxy table

To create a proxy table the steps are almost identical to create a scenario table. Select the type of proxy on the top of the sheet (if not already filled in for the scenario table) and press the button “Proxy table”. This will start the calculation engine of Cora and will show a progress window to the user. After a successful calculation the progress window will automatically closed, Notepad will open showing the log information. In this case **both** the scenario and proxy table will be written to the “Scenario results” sheet. This will always be on top of the previously calculated scenario or proxy table. For every type of proxy can only one scenario or proxy table be calculated, otherwise it will be overwritten. In the below image you will see an example of a scenario table and proxy tables.

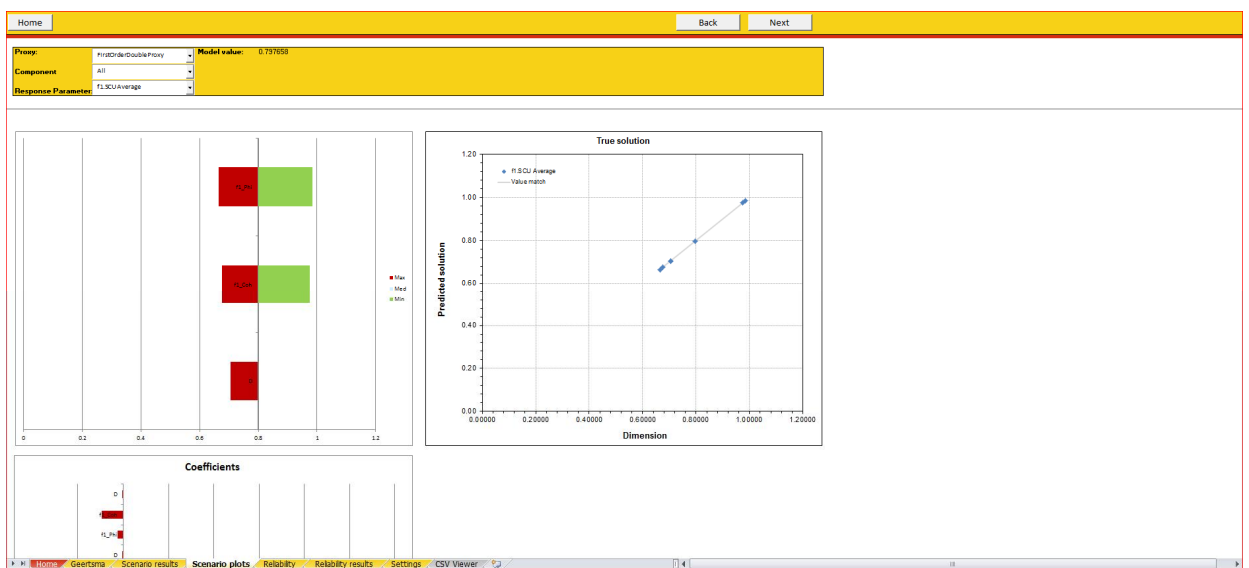
Home First order proxy P10 Scenario table Proxy table Back Next					
Current selected engine: Geertsma					
Scenario-FirstOrderProxy10					
	0	f1_Coh	f1_Phi		
P-10-50-50	1559.224051		3	16	
P-50-10-50	2000		1.718448061	16	
P-50-50-10	2000		3	10.87379225	
P-50-50-50	2000		3	16	
Response-FirstOrderProxy10					
	f1 SCU Average	f1 SCU Max	f1 SCU Area Fail m2	f1 SCU Area Fail %	Coefficient
P-10-50-50	0.703386949	0.731781295	0	0	0.34
P-50-10-50	0.97483822	1.032031573	170	0.4	0.34
P-50-50-10	0.984347408	1.036869591	200	0.4	0.34
P-50-50-50	0.797858036	0.84207926	0	0	0.34

The proxy table shows for each failure type the outcome of the engine for the given input in the scenario table. The example above shows that the depth doesn't have influence on the failed area (column f1.SCU Area Fail m2) but the strength parameters cohesion and angle of internal friction does have a lot of influence.

When you create a proxy table, a set of tables will write to the Excel sheet. Some of these tables will be used as data source for the scenario plots sheet. To navigate to the “Scenario plots” sheet please click the next button or just open the sheet by clicking the “Scenario plots” sheet tab.

9 Scenario plots

The scenario plots sheet is used to visualize the data that is generated from the different proxy tables. The three charts, Tornado, True solution and the Coefficients are populated with data from the “Scenario results” sheet. The three drop down boxes on the top of the sheet can be used to select the proxy and filter on the component and response parameter. Default the last created proxy will be displayed with no filter on components and the first available response parameter.

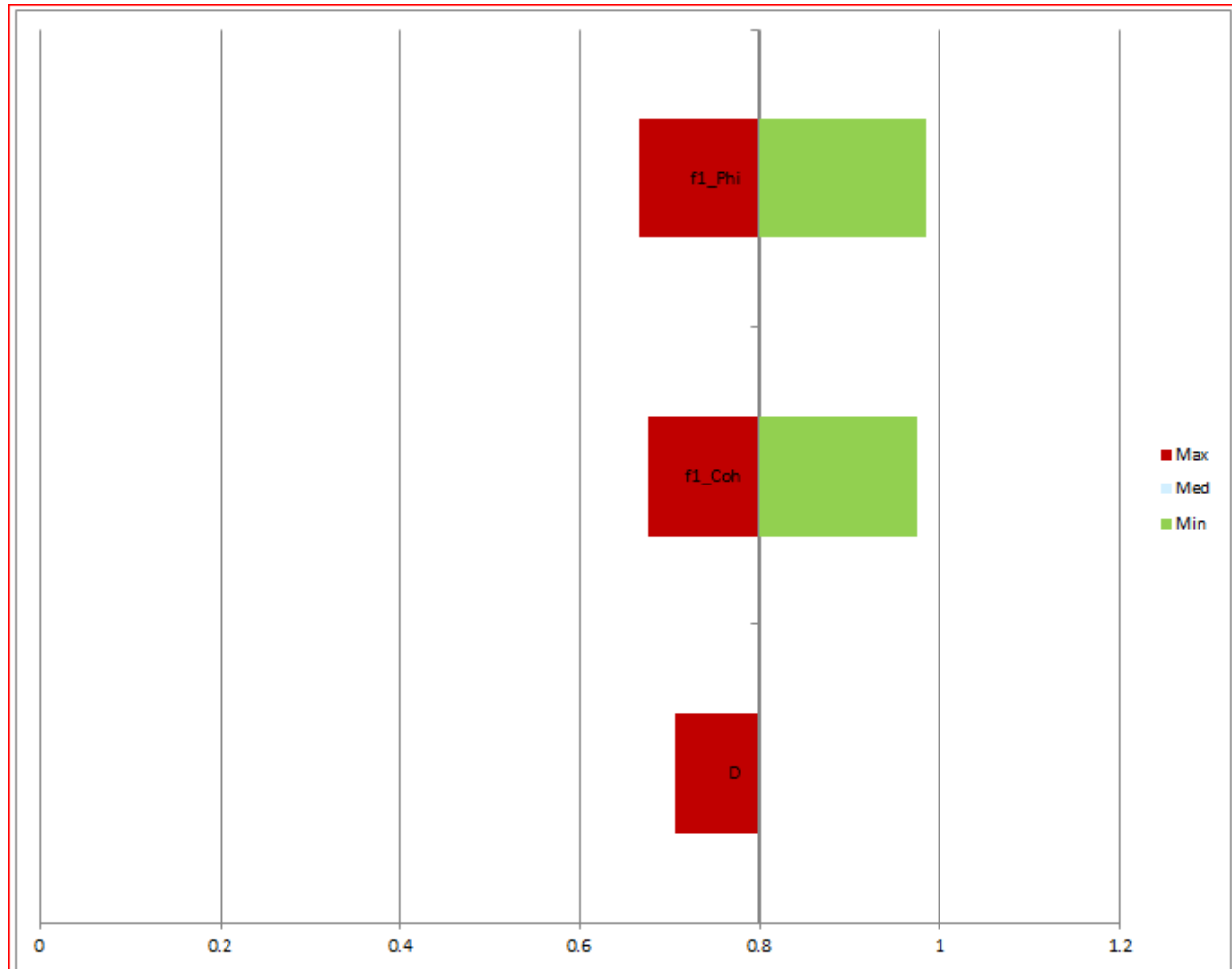


In the below image you see the three drop down boxes to filter the data that is visualized in the three different plots. The proxy drop down box is filled with all available proxy's that are generated for the current active engine. To add one you need to go back to the “Scenario results” sheet and create the missing proxy. Switching between the different proxy's will automatically update all the three plots on the “Scenario plots” sheet. When there is one or more component added to the active engine you can also filter on the component you would like to see. Based on the selected component the response parameter drop down box is populated with all available response parameters for that component. Switching the selected proxy table will also update the model value.

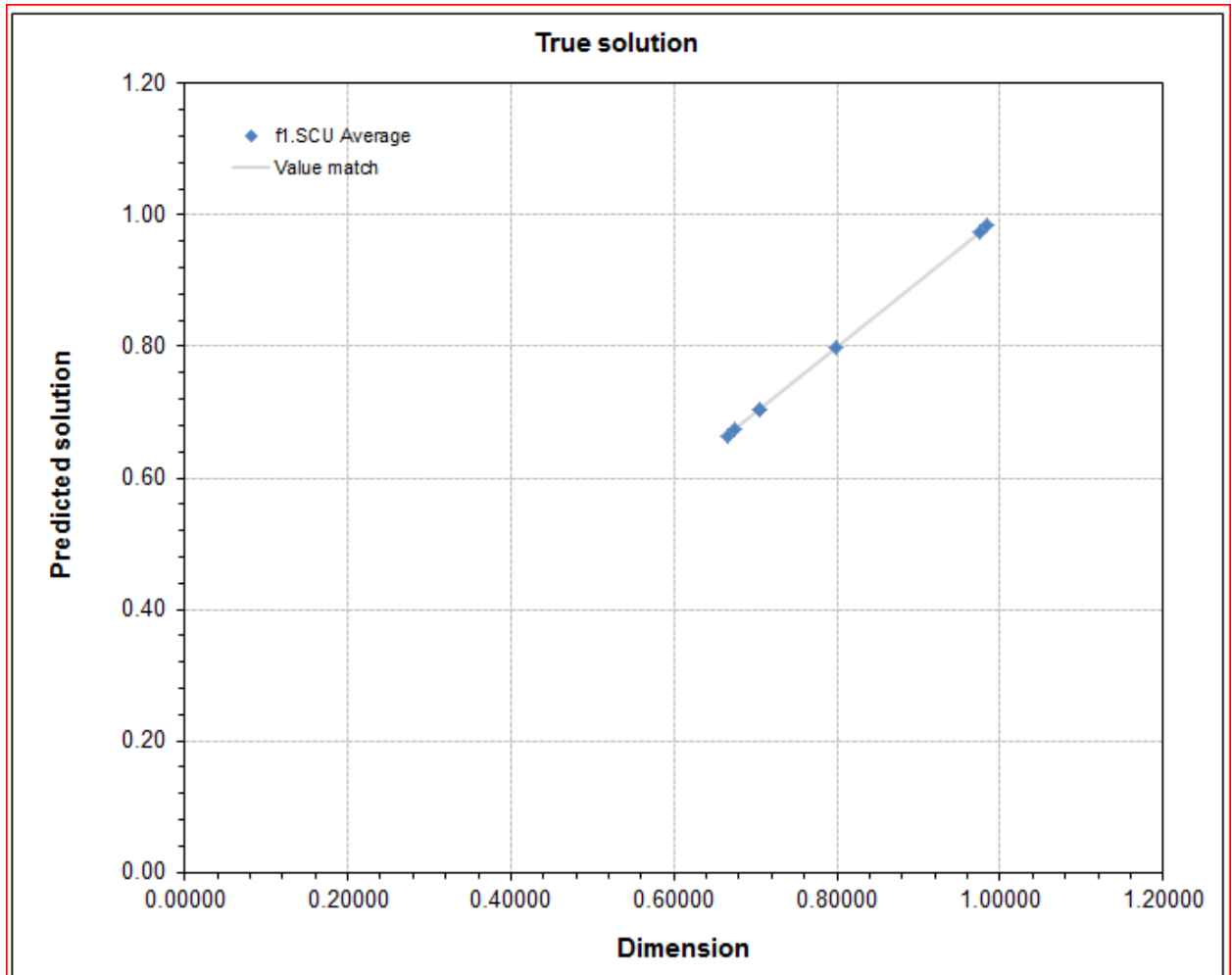
I suggest a short description of what is seen in the three graphs and what can be changed more, esp. the option components.

Proxy:	FirstOrderDoubleProxy	Model value:	0.797658036
Component	All		
Response Parameter:	f1.SCU Average		

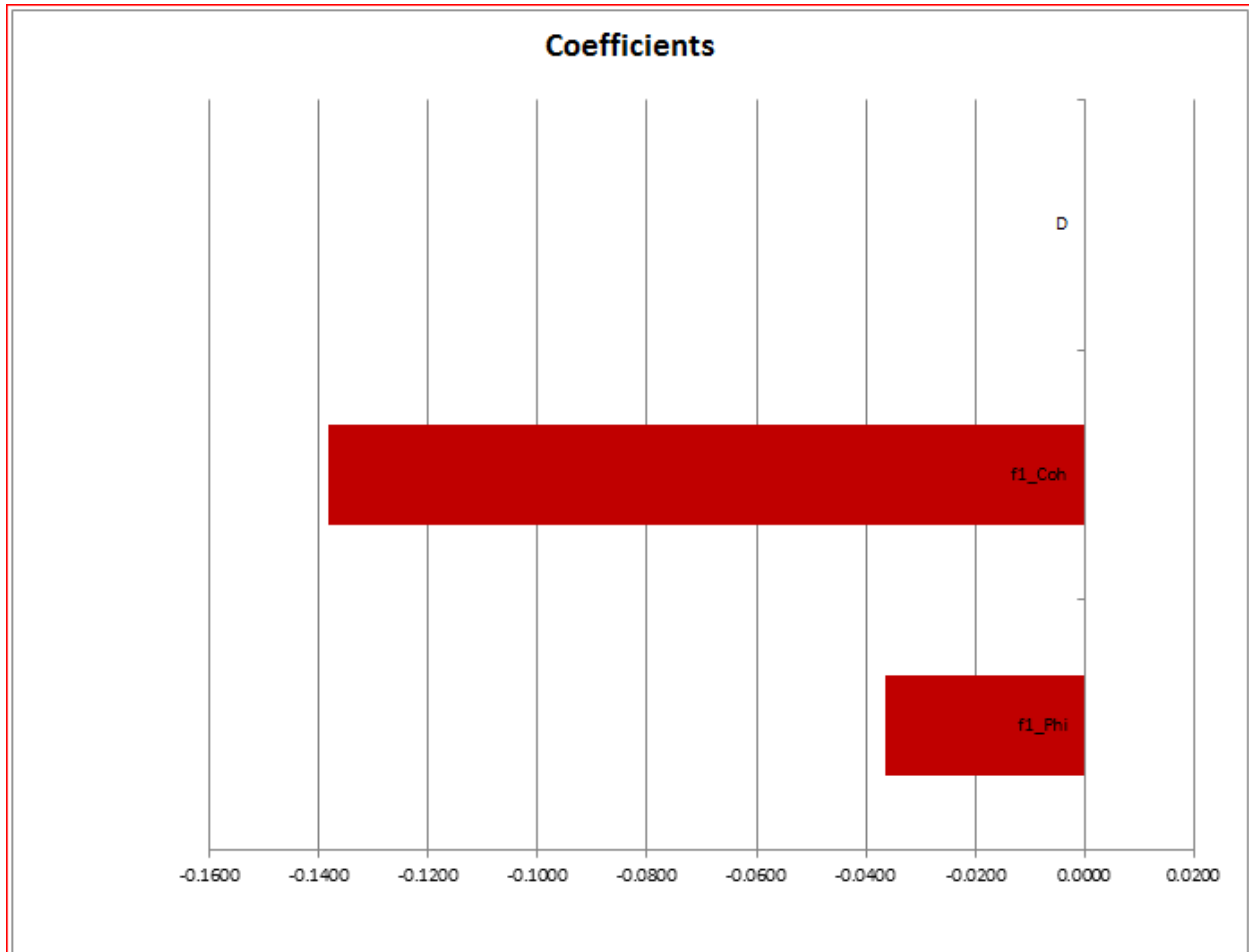
9.1 Tornado chart



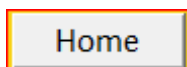
9.2 True solution chart



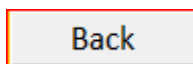
9.3 Coefficients chart



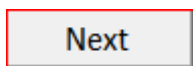
9.4 Navigation buttons



Button to navigate to the sheet named "Home".



Button to navigate to the previous sheet.



Button to navigate to the next sheet.

10 Reliability

To finally calculate the reliability you need to navigate to the “Reliability” sheet. There two ways to navigate to this sheet. You can open the sheet by clicking the tab “Reliability”, or click the next button on the “Scenario plots” sheet. In this sheet you must define the reliability calculation. This relates to the type of engine, the probabilistic method and the numerical parameters for that method and, of course, for each failure mechanism the value that is considered to be failure.

Based on the previous example where we used the “Geertsma” engine with a fault and shear failure, the “reliability” sheet will look like the below example

Home Run engine FORM Run engine

Current selected engine: Geertsma

Probabilistic configuration settings

Seed:	0	MinimumSamples:	1000	StartMethod:	StartZero	VarianceFactor:	-1
Umin:	-5	MaximumSamples:	10000	StartVector:		VariationCoefficientFailure:	0.1
Umax:	5	MaximumIterations:	50	RelaxationFactor:	0.75	VariationCoefficientNonFailure:	0.1
Intervals:	100					Threshold Relaxation Factor:	0.667

Select:	Failure Name:	Type:	Failure mechanism:	LSF Type:	Resistance:	Min:	Max:	Steps:
f1		Fault	Shear failure	SelectLSF	Resistance:			
f1		Fault	Shear failure	SelectLSF	Resistance:			
f1		Fault	Shear failure	SelectLSF	Resistance:			
f1		Fault	Shear failure	SelectLSF	Resistance:			

Home Geertsma Scenario results Scenario plots Reliability Reliability results Settings CSV Viewer Sheet4

10.1 Current selected engine

First of all it is important to check if the current engine is the engine you would like to do your reliability calculation on. The current selected engine can be found on the top of the sheet. To change this, go to the input sheet of the engine you would like to select and click the “Validate & continue” button. After this you can navigate manually to the “reliability” sheet, the current selected engine should now be changed.

Current selected engine:	Geertsma
--------------------------	----------

10.2 Probabilistic configuration settings

First you have to select the required probabilistic method in the right-hand-side drop-down box. The possible methods are presented in the final subsection of this Chapter

The Probabilistic configuration settings table can be divided in two sections. The top section is related to the selected probabilistic method and the bottom section is related to list of components for the selected engine. In the top section all probabilistic configuration parameter settings are listed, showed as below. Depending on the selected probabilistic method fields will turn green or yellow. Green fields are required input fields and can be changed, a default value will be filled in automatically when changing the probabilistic method.

Probabilistic configuration settings									
Seed:	0	MinimumSamples:	1000	StartMethod:	StartZero	VarianceFactor:	-1		
Umin:	-5	MaximumSamples:	10000	StartVector:		VariationCoefficientFailure:	0.1		
Umax:	5	MaximumIterations:	50	RelaxationFactor:	0.75	VariationCoefficientNonFailure:	0.1		
Intervals:	100					Threshold Relaxation Factor	0.667		

The cells that are green depend on the probabilistic method chosen. The meaning of the cells can be found in the report "CORA Design" report Deltares 1206709-0000-GEO-0074 January 2016.

On the bottom section of the table you will see a list of all the components for the active engine. For each of these components you will see the name, type and failure mechanism filled in for you. This will be repeated the number of LSF types that are available for the selected failure mechanisms. This will give the user the option to select multiple Limit state functions for the same component with the same failure mechanism. In the below image you will see how this would look like when there is one component of type fault with a failure mechanism Shear failure. None of them are selected yet.

Select:	Failure Name:	Type:	Failure mechanism:	LSF Type:	Resistance:	Min:	Max:	Steps:
	f1	Fault	Shear failure	Select LSF	Resistance:			
	f1	Fault	Shear failure	Select LSF	Resistance:			
	f1	Fault	Shear failure	Select LSF	Resistance:			
	f1	Fault	Shear failure	Select LSF	Resistance:			

To select one of the four rows you need to click on the first row in the green section and select the "x" or "X". After this you need to select the LSF type which stands for limit state function. Click in the same row on "Select LSF" and choose one of the LSF types from the dropdown box. At last you need to enter the resistance value; this can be one number or an array of numbers. Enter the resistance value on the same row for the min column, for an array also enter the max and step value. This can look like the below image:

Select:	Failure Name:	Type:	Failure mechanism:	LSF Type:	Resistance:	Min:	Max:	Steps:
X	f1	Fault	Shear failure	SCU Max	Resistance:	0.1		0.1
	f1	Fault	Shear failure	Select LSF	Resistance:			
	f1	Fault	Shear failure	Select LSF	Resistance:			
	f1	Fault	Shear failure	Select LSF	Resistance:			

All input is now in place to do a reliability calculation. This can be done by using directly the selected engine or by using the previously generated proxy tables.

10.3 Run engine

To calculate the reliability using the direct engine assuming that the engine sheet is correctly filled in, you need to do the following:

- Select “Run engine” in the first drop down box from the left on the menu bar.
- Select the type of probabilistic method in the second drop down box from the left on the menu bar.
- Click the “Run engine” button on the menu bar.

When there are no validations errors a progress window will show up. This window will show the estimated time and progress log as long as the engine is running. This window can be minimized so you can continue your work during the calculation.

After the calculation completed successfully the results will be showed in the “Reliability results” sheet.

This option uses for each drawing of the input variables a full engine calculation. The calculation time depends strongly on the time needed to do an engine calculation.

10.4 Run proxy

To calculate the reliability using a proxy table assuming that the engine sheet is correctly filled in, you need to do the following:

- Select “Run proxy” in the first drop down box from the left on the menu bar.
- Select the type of probabilistic method in the second drop down box from the left on the menu bar.
- Select the type of proxy in the third drop down box from the left on the menu bar. If there are no proxy tables available, please go to “Scenario results” sheet and create one or more proxy tables.
- Click the “Run proxy” button on the menu bar.

When there are no validations errors a progress window will show up. This window will show the estimated time and progress log as long as the engine is running. This window can be minimized so you can continue your work during the calculation.

After the calculation completed successfully the results will be showed in the “Reliability results” sheet.

This option is especially useful if engine calculations are time consuming. Often the proxies are a reasonable approximation of the required functions. Cora has the possibility to judge the accuracy of the proxies. If the accuracy is too low, an additional call to the engine is made. The validity of the proxies can be checked.

10.5 Probabilistic method table

Available probabilistic methods²:

Probabilistic methods
FORM
Monte Carlo
Importance Sampling
Directional Sampling

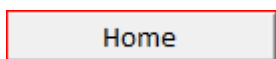
² See the design document for background on the probabilistic methods

11 Reliability results

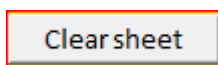
The reliability sheet is used to store all calculation results from the “Reliability” sheet. On the left side all input parameters are stored and on the right side the results. The results are expressed in the probability of failure due to the selected failure mechanisms.

Probabilistic configuration settings for Geertma calculated on 31/12/2014 5:19:10 PM										Reliability (1 Shear failure, SCU Max, Resistance=0.1) (FORM)		Design point: F1
Seed:	0	MinimumSamples:	1000	StartMethod:	ScanZero	VarianceFactor:	-1	F1 Shear failure, SCU Max				
Unit:	-5	MaximumSamples:	10000	StartVector:		VariationCoefficientFailure:	0.1	Beta	-35.10273632	0		
Omega:	5	MaximumIterations:	50	RelaxationFactor:	0.75	VariationCoefficientRoof Alfa	0.1	Probability of failure	1	1		
Intervals:	100			ThresholdRelaxationFactor:				Probability of non failure	2.07E-279	1		
Failure Name:	F1	Failure type:	Shear failure	Failure mechanism:	SCU Max	LSF Type:	Resistance	Threshold:	na			
							0.18, 2.0, 3.0, 4.0, 5.0, 6.0, 7.0, 8.0, 9.1					

11.1 Navigation buttons



Button to navigate to the sheet named “Home”.



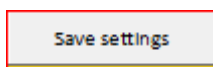
Button to clear all the calculation results on the sheet. To be handled with care, it is easy to remove hours of calculations.

12 Settings

All configuration setting of Cora can be found in the “Settings” sheet. Save changes pressing the “Save settings” button on the top of the page. These settings are stored to an external file on your computer; opening any other sheet will automatically load your settings to that sheet. This will prevent you from updating all the path settings when someone else is sending your there sheet.

			Save settings	
Sheet	Key	Value		
Cora	Path cora application	C:\Program Files (x86)\Deltares\CORA\Binaries		
QuickBlocks	Python path	C:\Program Files\QuickBlocks 2.1\python.exe		
QuickBlocks	QuickBlocks path	C:\Program Files\QuickBlocks 2.1\		
Cora	Verbosity	2		
GEOPAC	GEOPAC path	C:\Program Files\GEOPAC 4.2 - SVS		
GEOPAC	SaveAllCorruption	TRUE		
GEOPAC	SaveAllModelResults	TRUE		
GEOPAC	SaveModelResultsFailedCases	FALSE		
GEOPAC	SaveOutputFailedCases	TRUE		

12.1 Navigation buttons



Button to save your settings

A Working with a user defined model

This appendix describes how a model can be used within CORA. Such a model should fulfill some requirements, which are described in this appendix.

A.1 Environment

The model should have been developed in Python. The version supported is CPython (the “normal” version of Python). JPython, IronPython and Python for .Net are not supported. The Python environment must be available on your computer. You can specify its location in the CORA Excel sheet. From within the Python script, you can reference all kind of libraries, such as numpy or your own libraries.

Methods to be implemented

There are four methods which must be implemented:

Method: get_response

The `get_response` method calculates a response based on some input variables. This method contains your user defined model.

These input variables will be presented in the input form in the CORA Excel sheet. There you can give them a value or a stochastic distribution. The three other methods are needed to give a proper interaction with the Cora-routines.

The `get_response` method is supposed to perform a calculation and produces output. The input contains all variables that will be visible in Cora. The output must be returned as a one-dimensional array. This contains all response values calculated in your model. The order of these variables is up to you.

In case you need more input than the values passed in the method invocation, you can do this by reading a file or any other data source. Values read from such a data source will not be subject to probabilistic analysis and will not show in the CORA Excel sheet.

In case of errors, such as numerical instabilities or any other reason the response cannot be calculated, a not-any-number (nan) should be returned in the response array. Alternatively, you can raise an error in the Python way. Both ways will be treated the same by the probabilistic analysis: as a failed calculation from which the results will not be taken into account. Error messages will be logged in the CORA log file.

For debugging, you can use print statements in your code. They will be reflected in the CORA log file.

NOTE: the order of the variables is determined in the method `get_response` and must be kept in the following three methods.

Method: get_input_parameters

The `get_input_parameters` method is a parameterless method and returns the names of the input parameters. A string array must be returned. The size of the array should match the number of parameters in the `get_response` method.

Method: *get_response_parameters*

The *get_response_parameters* method is a parameterless method and returns the names of the response parameters as a string array. The size of this array must match the size of the array returned by the *get_response* method.

Method: *get_zvalue*

This method calculates the *z* value. The *z* value is an indication whether the calculation results (the response values) should be regarded as a failing system ($z < 0$) or not ($z \geq 0$). When using FORM as a probabilistic analysis method, the *z* function must be continuous with respect the input values.

This method has three input parameters, being a string value corresponding with a response parameter, the response (produced by the *get_response* method) and a resistance. The last value is a value you can enter in the CORA Excel sheet and use to your own preference. A double value (the *z* value) should be returned.

In case of errors, the same techniques are available as in the *get_response* method.

A.2 Example 1

The following code is a simple, non-realistic example.

```
import sys
import numpy as np

def get_response(par1,par2,par3,par4,par5):
    res1 = par5 - par4 - par3 - par2 - par1
    print "Result 1 = ", res1
    res2 = par5 + par4 - par3 - par2
    return [res1, res2]

def get_input_parameters():
    return ["A", "B", "C", "D", "E"]

def get_response_parameters():
    return ["Result 1", "Result 2"]

def get_zvalue(responsename, response, resistance):
    if responsename == "Result 1":
        z = resistance - response[0]
    elif responsename == "Result 2":
        z = resistance - response[1]
    else:
        raise ValueError("response name unknown")
    return z
```

A.3 Example 2

As a second example we consider the question to do an analysis of a block with mass *m* on a sloping flat surface with slope angle α and friction ratio μ . Also a parallel force *F* is acting in upward direction to the mass.

We can calculate the normal force $N = mg \cos \alpha$ and shear force $S = mg \sin \alpha$. The system fails if the block starts to slide. We define two response values (although this is not the cleverest way from mathematical point of view, but we want to have two failure values):

- $z_1 = \mu N + S - F = \mu mg \cos \alpha + mg \sin \alpha - F$ (for upward motion).
- $z_2 = \mu N - S - F = -\mu mg \cos \alpha + mg \sin \alpha - F$ (for downward motion).

```
# import sys
import numpy as np
import myownlib

def get_response(m, mu, alpha, F):
    # from a library or read from an ini-file etc.
    g = myownlib.gravity
    z1 = mu * m * g * np.cos(alpha) + m * g * np.sin(alpha) - F
    z2 = -mu * m * g * np.cos(alpha) + m * g * np.sin(alpha) - F
    return [z1, z2]

def get_input_parameters():
    return ["mass", "friction ratio", "slope angle", "Driving force"]

def get_response_parameters():
    return ["upward motion", "downward motion"]

def get_zvalue(responsename, response, resistance):
    if responsename == "upward motion":
        z = resistance - response[0]
    elif responsename == "downward motion":
        z = resistance - response[1]
    else:
        raise ValueError("response name unknown")
    return z
```